Sri Krishna Prabhas Yadla

ee23btech11224@iith.ac.in | srikrishnaprabhas12@gmail.com | +91 7893836752 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.29
FIITJEE Junior College	Class XII (TSBIE)	2023	96.40%
FIITJEE School	Class X (SSC)	2021	10.00%

SCHOLASTIC ACHIEVEMENTS

Director's Academic Excellence Awardee, IITH; Department Topper 2023, 2025

NTSE 2020-21: Secured rank **6** in stage 1

EXPERIENCE

Physics-Informed Neural Network for Semiconductor Electrostatics

Research Project under Dr.Oves Badami | IITH

- Developed a physics-informed neural network **(PINN)** for **1-D** electrostatic potential modeling of double-gate **oxide-semiconductor-oxide** devices across **3-40 nm** channel lengths and **0.6-1.2 V gate bias.**
- Integrated **Poisson's equation, oxide-silicon interface flux continuity, and supervised conduction-band data** into a unified training objective, enforcing physical constraints beyond data fitting.
- Introduced a **trainable Fermi level** to jointly model electrostatic potential and carrier statistics.
- Implemented **GradNorm adaptive loss balancing** with normalized losses and periodic weight updates to mitigate severe scale imbalance between PDE, interface, and data losses.
- Used **automatic differentiation and second-order derivatives** to evaluate Poisson residuals, with double precision and smooth **xtanh(x) activation** - Optimized network/training hyperparameters using **Optuna.**
- Achieved **4.47-8.74 mV worst-case MAE** across 0.6-1.2 V gate bias while reducing PDE and interface physics residuals by 4 orders of magnitude versus data-only training.

SKILLS

- **Programming Languages:** C, C++, Verilog, Python, Bash, ARM
- **Software:** Git, LaTeX, Solid Edge (CAD), MATLAB
- **Libraries/Frameworks:** NumPy, Pandas, Matplotlib, Pytorch, Jupyter

PROJECTS

3D Game Console using FPGA

- Designed an application-specific **GPU** in **Verilog** for real-time rendering on a **400×240 LCD.**
- Implemented **parallel** triangle **rasterization** with **depth testing** and **Z-buffering** in FPGA hardware
- Optimized **rasterization** by migrating compute-intensive routines from **custom ISA assembly** to **Verilog**
- Partitioned the graphics pipeline between **C-based vertex processing** and **FPGA hardware** for acceleration
- Developed a C++ game engine **with event-driven game-object** and **trigger management** for gameplay
- Integrated **gyroscope controls** with the **FPGA graphics pipeline** for real-time interactive rendering

Handwritten OCR and Sentiment Analysis without Pretrained Models

- Trained a custom **CNN** to classify handwritten characters (A-Z) using a grayscale image dataset.
- Developed an end-to-end OCR-to-NLP pipeline that converts handwritten text into digital form and performs sentiment analysis.
- Implemented sentiment classifier on extracted text using **Naive Bayes** method.

Classification and Geospatial Visualization of Pincode Data

- Filtered and cleaned India Post dataset to isolate Telangana region entries using latitude, longitude.
- Visualized spatial distribution of valid pincodes using custom map plots with **matplotlib**.
- Implement the **k-means++ clustering** algorithm from scratch to identify distinct clusters within Telangana.

Programmable, Selective Noise Canceller

- Developed **noise cancellation** system with both **full** and **partial** suppression modes in MATLAB.
- Implemented Recursive Least Squares (**RLS**) algorithm achieving a **45 dB SNR gain**, outperforming LMS/NLMS methods.
- **Designed** a programmable **IIR notch filter** for partial suppression **to retain user-selected tonal frequencies.**
- **Optimized** the adaptive filter architecture **by evaluating convergence, SNR, latency, and buffer-size trade-offs.**

RELEVANT COURSES

Computer Science: Introduction to Programming, Data Structures and Applications, Linear Optimization

Math: Applied Statistics, Probability, Matrix Theory, Predictive Modeling, Time Series Forecasting, Combinatorics and Graph Theory, Calculus, Differential Equations, Complex Variables, Numerical Methods for ODE

Digital: Digital Systems, Microprocessor & Computer Architecture, RTL Design & Verification, Digital IC Design, Intro to VLSI

POSITIONS OF RESPONSIBILITY

Teaching Assistant for **Matrix Theory** under Prof. GVV Sharma

Core, Elektronica - Electronics and Signal Processing club of IIT Hyderabad

Teaching Assistant for **Waveguides and Antennas** under Prof. Aneesh Shobanan

EXTRACURRICULAR

Hobbies & Interests: Acrylic Painting & Drawing, Sketching, Reading

NCC 'B' certificate Holder